Agentic Visual Perception in Distributed Autonomous Systems

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Distributed autonomous systems let teams of drones and robots search large areas, inspect infrastructure, and respond to incidents from several viewpoints. This capstone focuses on cooperative search-and-inspect missions in which visually equipped agents share only the information needed to coordinate. Development will begin in simulation, followed, if feasible, by a small physical demonstration.

The hard part is deciding when an uncertain visual report should change the mission. A detection may be stale, incomplete, duplicated, or contradicted because of occlusion, sensing error, object motion, or a changed scene. Acting too soon can close the wrong task; verifying every report wastes time. Commercial programs address adjacent problems: Skydio documents cloud-coordinated multi-drone deconfliction for inspection and incident response, Anduril coordinates diverse autonomous systems, and Shield AI develops collaborative autonomy for multiple aircraft under human supervision.

We propose to build and evaluate an evidence-gated visual task-orchestration framework. Each observation becomes a structured claim recording its source, time, pose, model, confidence, and visual artifact. These claims would update revisable entity beliefs, and explicit policies then create, verify, close, reopen, or reassign tasks. Agents would receive task-scoped states instead of raw global data, and a causal trace would record why each decision occurred. Tests would compare the framework with first-report, majority-vote, and always-verify baselines under controlled disagreement, staleness, occlusion, and agent unavailability. Simulator truth is reserved for scoring, while navigation and flight control remain outside the framework.